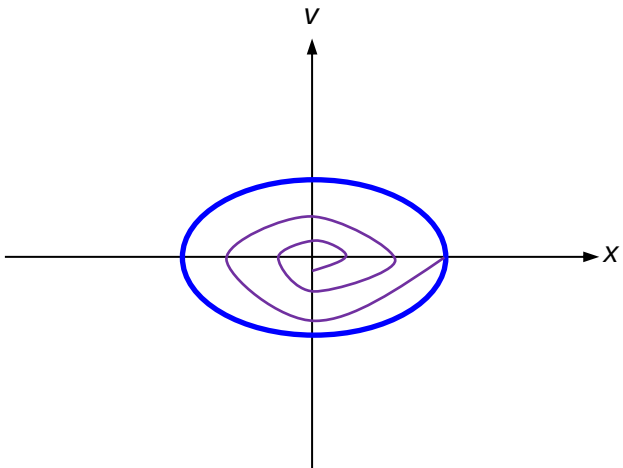


3	
(a)	As the tube is partially submerged, there is upthrust.
	The weight and upthrust are equal in magnitude and acts in opposite directions. Hence, the resultant force is zero.
(b)	At equilibrium, weight of the tube and sand = upthrust $Mg = \rho Ahg$
	When further displaced and taking downwards as positive, $Mg - \rho A(h + x)g = Ma$
	$\rho Ahg - \rho Ahg - \rho Axg = Ma$ $-\rho Axg = Ma$ $a = -\left(\frac{\rho Ag}{M}\right)x$
(c)	Since ρ , A , g and M are constants, acceleration is directly proportional to displacement.
	The negative sign shows that acceleration is in the opposite direction to displacement.

(d)	$\omega^2 = \frac{\rho Ag}{M}$
	$\frac{2\pi}{T} = \sqrt{\frac{\rho Ag}{M}}$ $f = \sqrt{\frac{\rho Ag}{4\pi^2 M}}$ $f = \sqrt{\frac{(1.2 \times 10^3)(5.5 \times 10^{-4})(9.81)}{4\pi^2 (120 \times 10^{-3})}}$
	$= 1.2 \text{ Hz (2 s.f.)}$
	
(e)(i)	Correct shape.
(e)(ii)	Spiral inwards clockwise starting from positive maximum displacement (downwards positive).
4	